SYSDESIGN REFERENCE

Eventual Consistency Tradeoff Matrix

A decision-oriented comparison of alternatives and failure modes.

Allow replicas to converge after writes while defining what users can safely observe meanwhile.

VERSION 1.0 REVIEW

REVIEWED THROUGH 2026-07-18

STATUS HUMAN REVIEW REQUIRED

SYSDESIGN / EVENTUAL-CONSISTENCY

SYSDESIGN

Eventual Consistency Tradeoff Matrix

CORE IDEA

Allow replicas to converge after writes while defining what users can safely observe meanwhile.

Reference architecture decision chain

01	02	03	04
Conflict Resolution	Read-Your-Writes	Versioning	Reconciliation

Decisions to defend

+ Conflict resolution
+ read-your-writes
+ versioning
+ reconciliation
+ and UX states.

Failure modes to test

+ Lost updates
+ oscillating views
+ and irreversible actions based on stale reads.

Interview flow

- [ ] Clarify users, critical paths, and non-goals
- [ ] Estimate only scale that changes a decision
- [ ] Define API, data model, and source of truth
- [ ] Draw synchronous and asynchronous paths
- [ ] Pressure-test timeout, retry, overload, and recovery
- [ ] Close with tradeoffs and one alternative

Deep-dive drill

Eventual Consistency Tradeoff Matrix

Probe	What a strong answer makes explicit
How does the design behave during a regional dependency failure?	Requirement, decision, cost, failure behavior, and observable metric.
Which metric reveals the first bottleneck?	Requirement, decision, cost, failure behavior, and observable metric.
What consistency does the user actually observe?	Requirement, decision, cost, failure behavior, and observable metric.
How would the design change at ten times the traffic?	Requirement, decision, cost, failure behavior, and observable metric.

Data questions

+ Partition key
+ Versioning
+ Uniqueness
+ Retention
+ Reconciliation

Operations questions

+ SLO
+ Backlog
+ Capacity
+ Alert
+ Failover test

DESIGN DISCIPLINE

For every box: which requirement justifies it, what cost does it introduce, and what happens when it fails?

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